

5-Day Pathogen (Bacterial) Genomics Workshop Agenda

Time	Type	Session	Facilitator
DAY 1 — Foundations of Bacterial Genomics & Sequencing Quality Control			
08:30 – 09:00	Plenary	Welcome & Workshop Overview	<i>Shahiid Kiyaga</i>
09:00 – 10:30	Theory	Basics of Bacterial Genomes & Genome Organization	<i>Shahiid Kiyaga</i>
10:30 – 11:00	Break	Tea / Coffee Break	
11:00 – 12:30	Theory	Quality Control of Sequencing Data	<i>Shahiid Kiyaga</i>
12:30 – 13:30	Break	Lunch Break	
13:30 – 15:00	Practical	Practical: Quality Control Assessment	<i>Shahiid & Patricia</i>
15:00 – 15:30	Break	Tea / Coffee Break	
15:30 – 17:00	Practical	Practical: Trimming Fastq	<i>Shahiid & Patricia</i>

DAY 2 — Genome Assembly, Genome Alignment & Variant Calling			
08:30 – 09:00	Plenary	Day 1 Recap & Q&A	<i>Shahiid Kiyaga</i>
09:00 – 10:30	Theory	Assembly Principles: Reference vs De Novo	<i>Shahiid Kiyaga</i>
10:30 – 11:00	Break	Tea / Coffee Break	
11:00 – 12:30	Theory	Genome Alignment & Variant Calling	<i>Shahiid Kiyaga</i>
12:30 – 13:30	Break	Lunch Break	
13:30 – 14:30	Practical	Practical: <i>Denovo</i> Genome Assembly	<i>Shahiid & Patricia</i>
14:30 – 15:00	Break	Tea / Coffee Break	
15:00 – 17:00	Practical	Practical: Genome Alignment & Variant Calling	<i>Shahiid & Patricia</i>

DAY 3 — Genome Annotation & Pan-genome Analysis			
08:30 – 09:00	Plenary	Day 2 Recap & Q&A	<i>Shahiid Kiyaga</i>
09:00 – 10:30	Theory	Bacterial Genome Annotation	<i>Shahiid Kiyaga</i>
10:30 – 11:00	Break	Tea / Coffee Break	
11:00 – 12:30	Theory	Pan-genome Analysis & Multilocus Sequence Typing	<i>Shahiid Kiyaga</i>
12:30 – 13:30	Break	Lunch Break	
13:30 – 15:00	Practical	Practical: Annotation and Pan-genome Analysis	<i>Shahiid & Patricia</i>
15:00 – 15:30	Break	Tea / Coffee Break	
15:30 – 17:00	Practical	Practical: MLST Database Queries	<i>Shahiid & Patricia</i>

DAY 4 — Phylogenetics, AMR, Virulence & Mobile Genetic Elements			
08:30 – 09:00	Plenary	Day 3 Recap & Q&A	<i>Shahiid Kiyaga</i>
09:00 – 10:30	Theory	Mechanisms of AMR & Virulence	<i>Shahiid Kiyaga</i>
10:30 – 11:00	Break	Tea / Coffee Break	
11:00 – 12:30	Theory	Horizontal Gene Transfer & Mobile Genetic Elements	<i>Shahiid Kiyaga</i>
12:30 – 13:30	Break	Lunch Break	
13:30 – 14:30	Practical	Practical: AMR, Virulence & Plasmids	<i>Shahiid & Patricia</i>
14:30 – 15:00	Break	Tea / Coffee Break	
15:00 – 17:00	Practical	Practical: Phylogenetic Analysis	<i>Shahiid & Patricia</i>

DAY 5 — Applications, Visualization & Group Project Presentations			
08:30 – 09:00	Plenary	Day 4 Recap & Q&A	<i>All Facilitators</i>
09:00 – 10:30	Theory	Applications of Bacterial Genomics: AMR Surveillance	<i>Shahiid Kiyaga</i>
10:30 – 11:00	Break	Tea / Coffee Break	
11:00 – 13:00	Practical	Group Project: Final Analysis & Presentation Preparation	<i>All Facilitators</i>